STAFF REVIEW · H446 219 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. The unsettled vertex with the smallest current tentative distance.

2. The cost of travelling between two connected vertices.

3. The previous vertex on that route.

4. Candidate 3+4=7, so update B to 7 and predecessor(B)=A.

I Do — source worked steps

1. Predict the next state — Start S=0: tentative A=2, B=6.

2. Update the table — Settle A=2; B improves to 4 and T receives 9.

3. Update the table — Settle B=4; T improves to 5.

4. Update the table — Trace T←B←A←S, then reverse: S→A→B→T at cost 5.

We Do — guided CHECK

1. Update the table
R improves Q to 3 and gives T=11.
Q improves T to 6.
T←Q←R←P reverses to P→R→Q→T, cost 6.

You Do — source mark schemes

1. [4] U→W costs 1 [1]; W→V adds 2 [1]; total 3 [1]; it is shorter than the direct cost 4 [1].

2. [6] Start B=1, A=5 [1]; settle B=1 [1], improve A=3 [1]; settle A=3, T improves to 6 [1]; route S→B→A→T [1]; predecessor(T)=A [1].

3. [6] Z is unreachable [1], remains at ∞ [1] and has no valid predecessor [1]; once all unsettled vertices have ∞ distance [1], no reachable vertex remains [1] and processing can terminate [1].

Exit Ticket — source answer

1. True. The predecessor of each node gives the step immediately before it on the path.